

Lecture 01

Introduction to Environmental Economics

Ivan Rudik
AEM 4510

Roadmap

- Course set up
- What is environmental economics?
- What are the goals for this class?
- Microeconomics recap

Course set up

AEM 4510 / AEM 5510 / ECON 3865: Basics

Professor: Ivan Rudik

TAs: Chrisma D'Souza, Nathan Bol, Becca Ford

Office hours: see canvas

Readings: Markets and the Environment by Keohane and Olmstead, papers available through the course website or library

Prereqs: MATH 1110, and either AEM 2600 or ECON 3030

Course website: <https://aem4510.ivanrudik.com>

AEM 4510 / AEM 5510 / ECON 3865: Grading

5 prelims (20% each): canvas-based, simple questions (solo)

- 5 in class prelims + final: lowest score is dropped
- Prelims will be a combination of multiple choice, short answer, graphical, and simple mathematical questions that will not require a calculator

3 Problem Sets (15%, 10%, 5%): mix of quantitative and conceptual problems to give you practice in learning key concepts (teams of 3)

- 1 before each prelim: Feb 6, Mar 13, Apr 17

AEM 4510 / AEM 5510 / ECON 3865: Grading

3 Prelims (15%, 15%, 10%): test your understanding of key concepts (solo)

- Feb 15, Mar 22, Apr 26
- **SDS:** This class is in the Alternative Test Program, if approved for exam accommodations, you must request your exam accommodation(s) for this course and fill out an exam request form for each exam in this course via the SDS student portal by February 8th

1 Final Project (5% paper, 2% presentation, 3% team eval): put your understanding of key concepts to work on real world issues (teams of 3)

- Presentations last 3ish days of class, paper due May 16 during finals week

AEM 4510 / AEM 5510 / ECON 3865: Content

Weeks 1-7ish: the economics of environmental regulation

- More theoretical, graphical
- Taxes, cap and trade, regulating monopolies, etc

Weeks 7-14ish: using markets (real and financial) to tell us about the environment

- Some theory, also real world analyses and data
- valuing recreation, housing markets, bond markets, green instruments

AEM 4510 / AEM 5510 / ECON 3865: Last words

This was just an overview, syllabus on canvas and course website has a lot more details about specific requirements

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Any questions about the course?

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These ideas can be applied to the environment

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How agents will response to climate change policies

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The value of mitigating pollution

How agents will response to climate change policies

Whether investment tax credits for wind power are cost-effective

Air pollution is bad



How do people respond to info?



Sequoia Kings Air
@SequoiaKingsAir

ADVISORY: #ozone is expected to be Unhealthy for All today in and around the foothills area of @SequoiaKingsNPS (Ash Mountain entrance)

Ozone Air Quality Guide

Advisory Level	Health Concern
Unhealthy (86 ppb – 105 ppb)	The following groups should avoid prolonged outdoor exertion: • People with lung disease, such as asthma • Children and older adults • People who are active outdoors Everyone else should limit prolonged outdoor exertion.

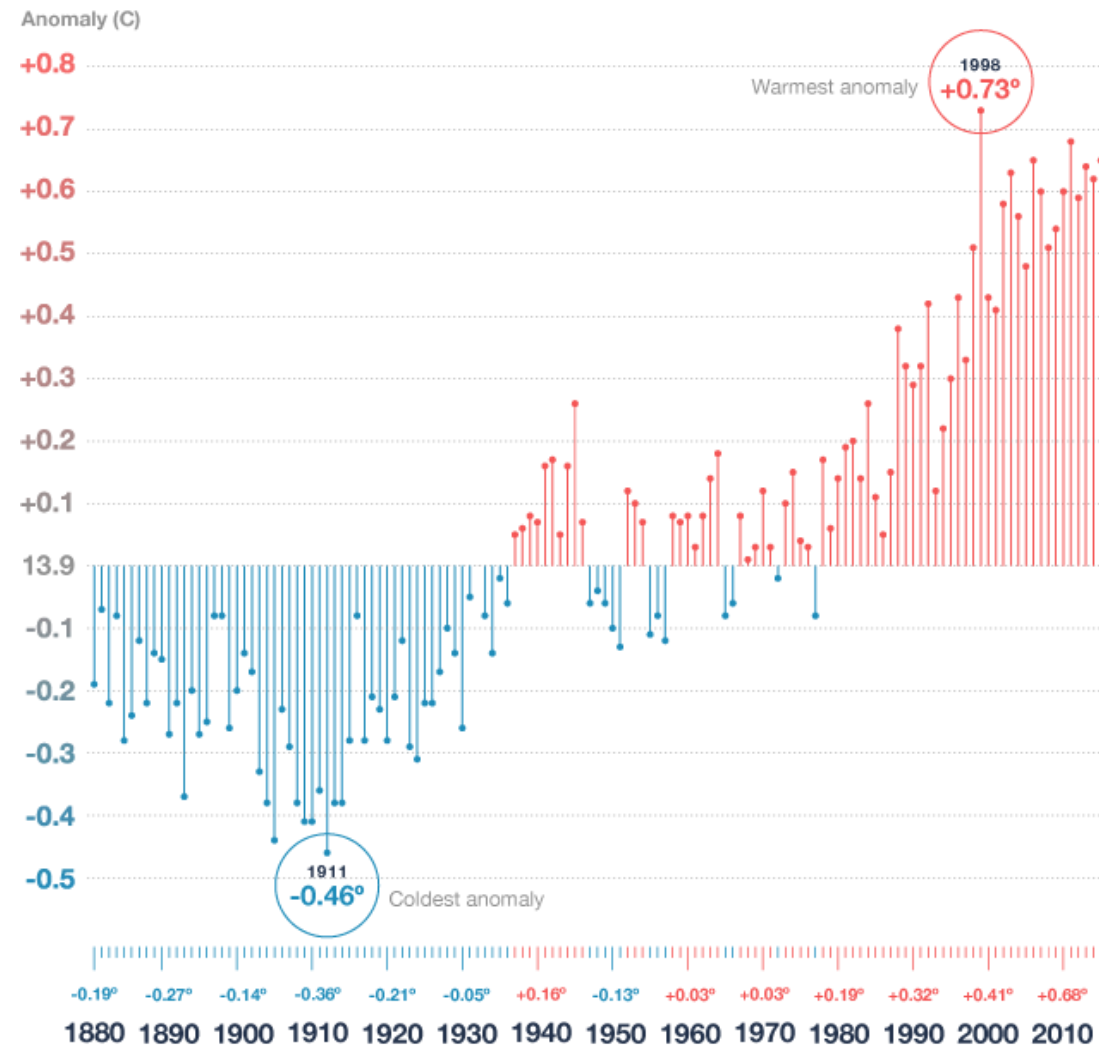
The ozone advisory level is based on an 8 hour average ozone concentration; ppb = parts per billion.

Places provide info to help people avoid air pollution

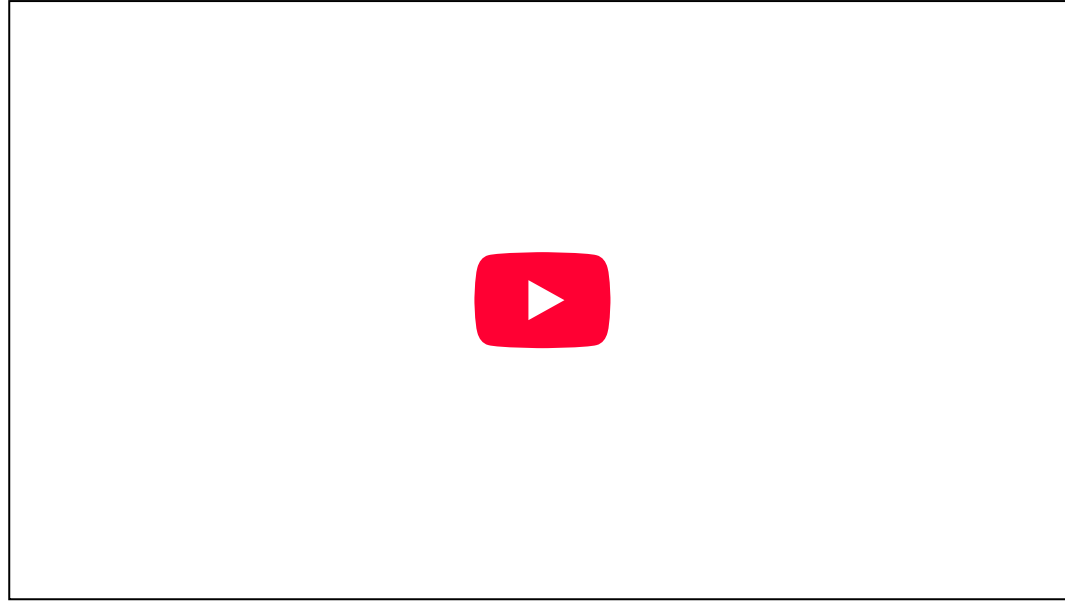
Does it work?

How well?

Climate change



Climate change



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Why do economists care about climate change?

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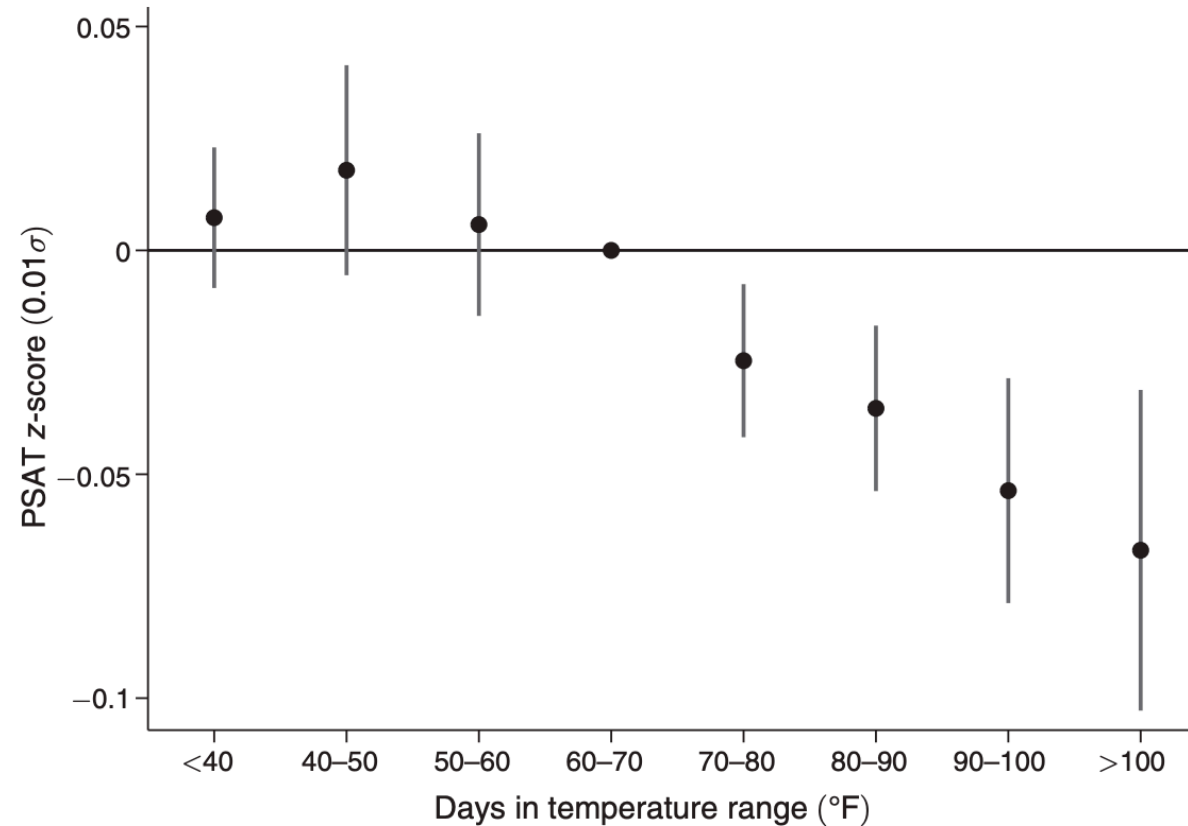
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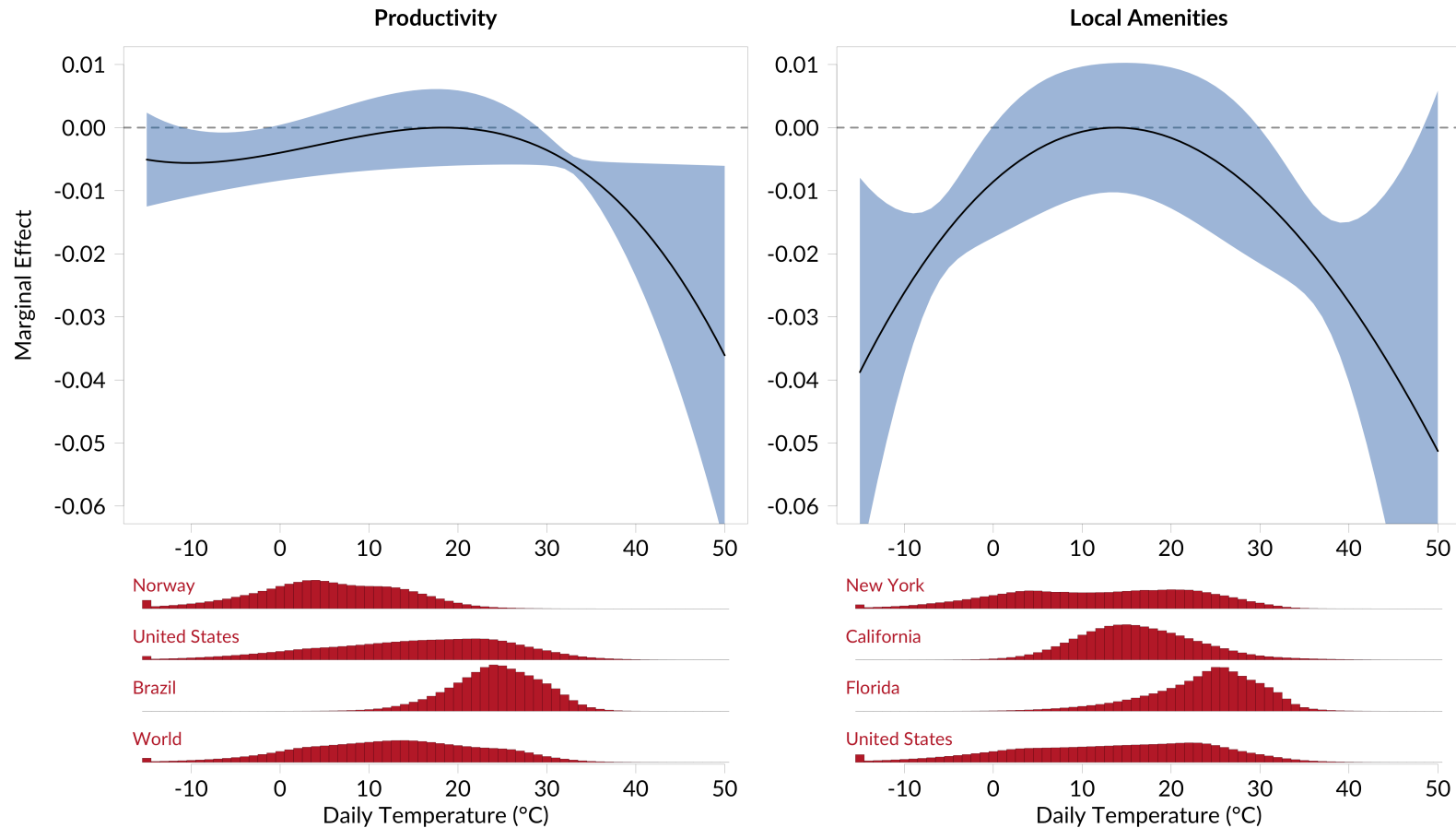
- Production
- Learning
- Leisure
- Fishing
- etc, etc

Climate change: heat hurts learning



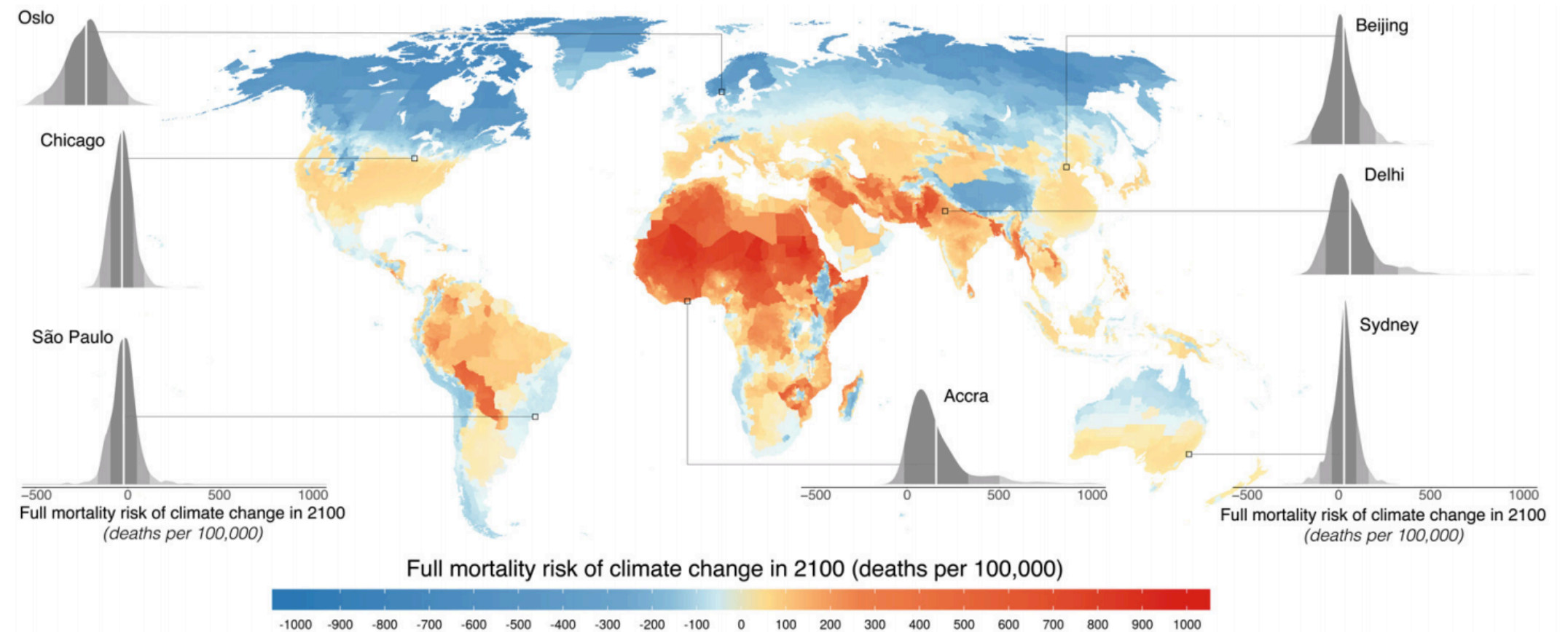
Park et al. (2020)

Climate change: heat hurts the economy



Rudik et al. (2021)

Climate change: extreme heat/cold increases mortality

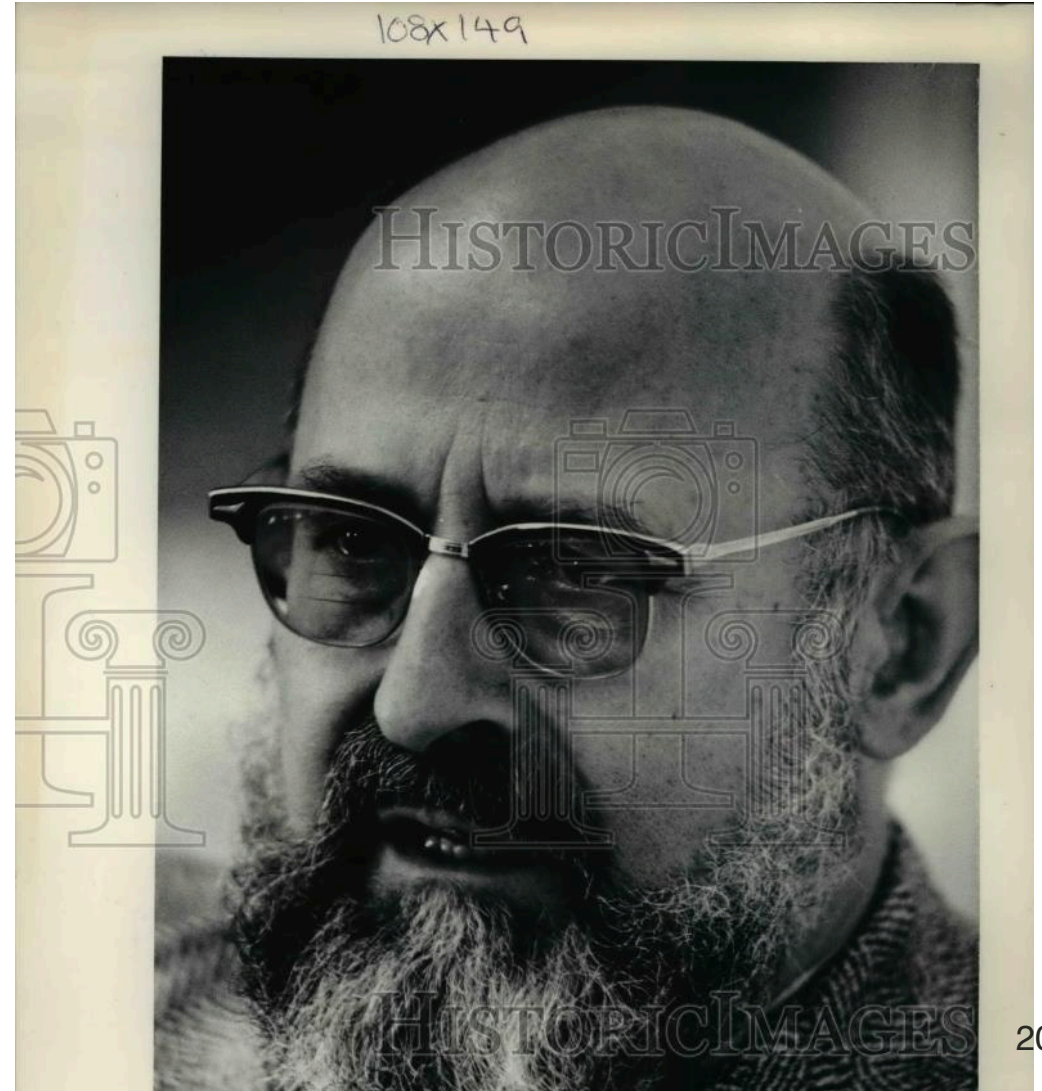


What is enviro econ?

Environmental economics is actually pretty new

Spurred by **John Krutilla** in the 1950s

His paper **Conservation Reconsidered** is the landmark paper in the field (sort of like *Wealth of Nations* and economics as a whole)



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Prepare you for after college:

- Industry jobs (energy, transportation, finance)
- Public sector and NGOs (EPA, DOE, RFF, Brookings, Federal Reserve)
- Graduate programs

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Learn both the theory and **applications** of environmental economics

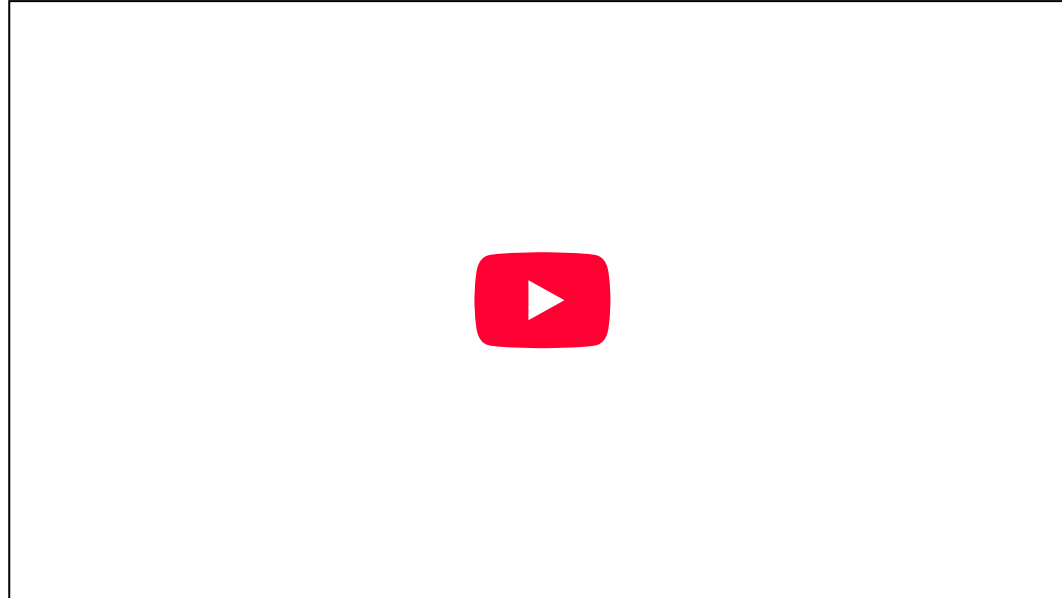
Microeconomics recap

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Is Intro to Micro applicable everywhere?

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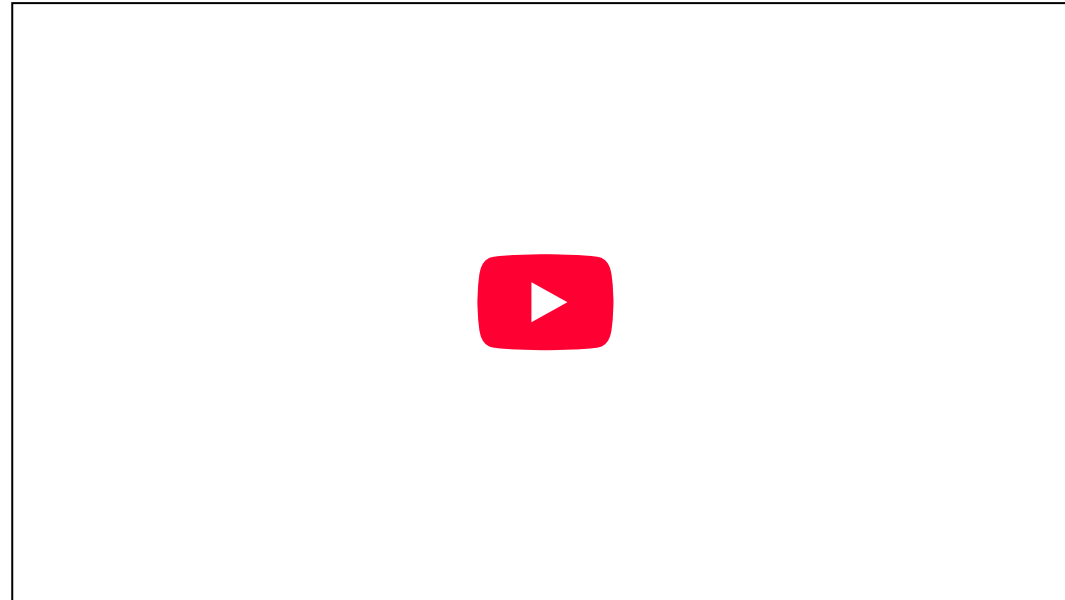


Creating markets to solve problems

How do we solve some problems in practice?

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Micro recap: terminology

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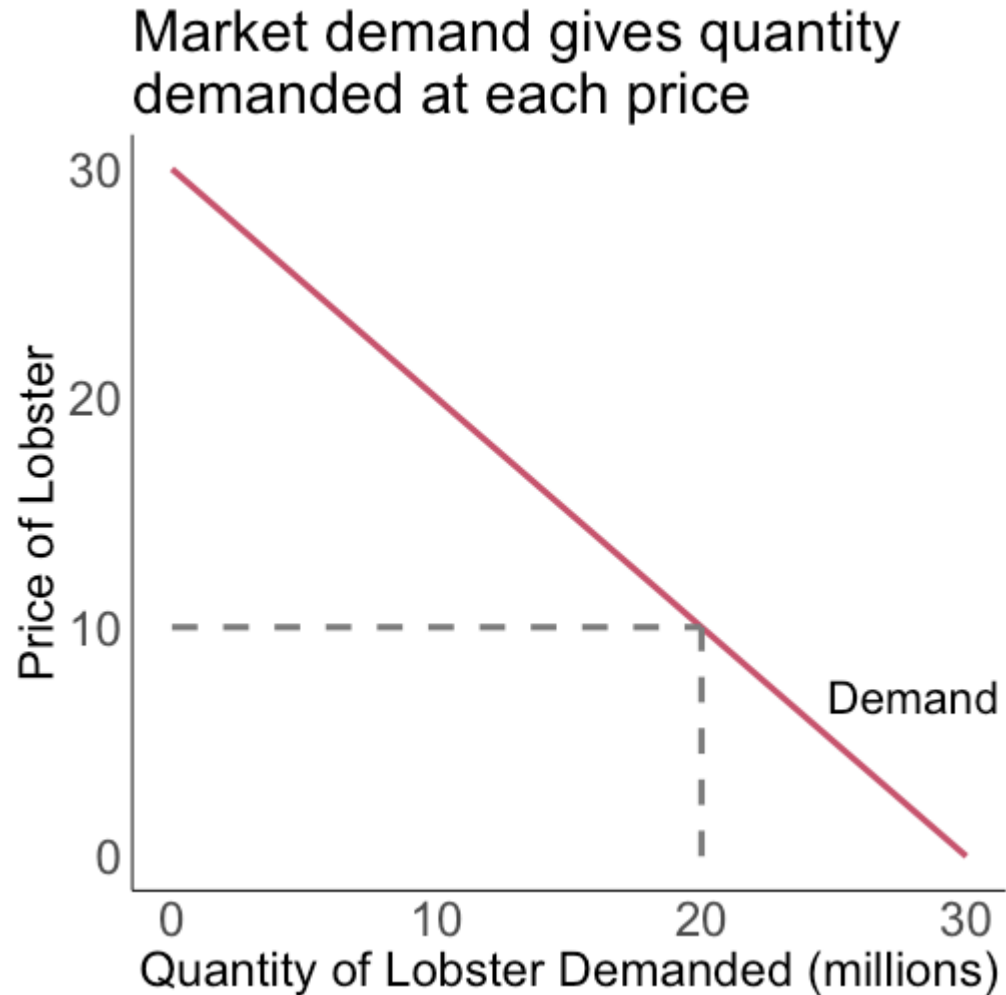
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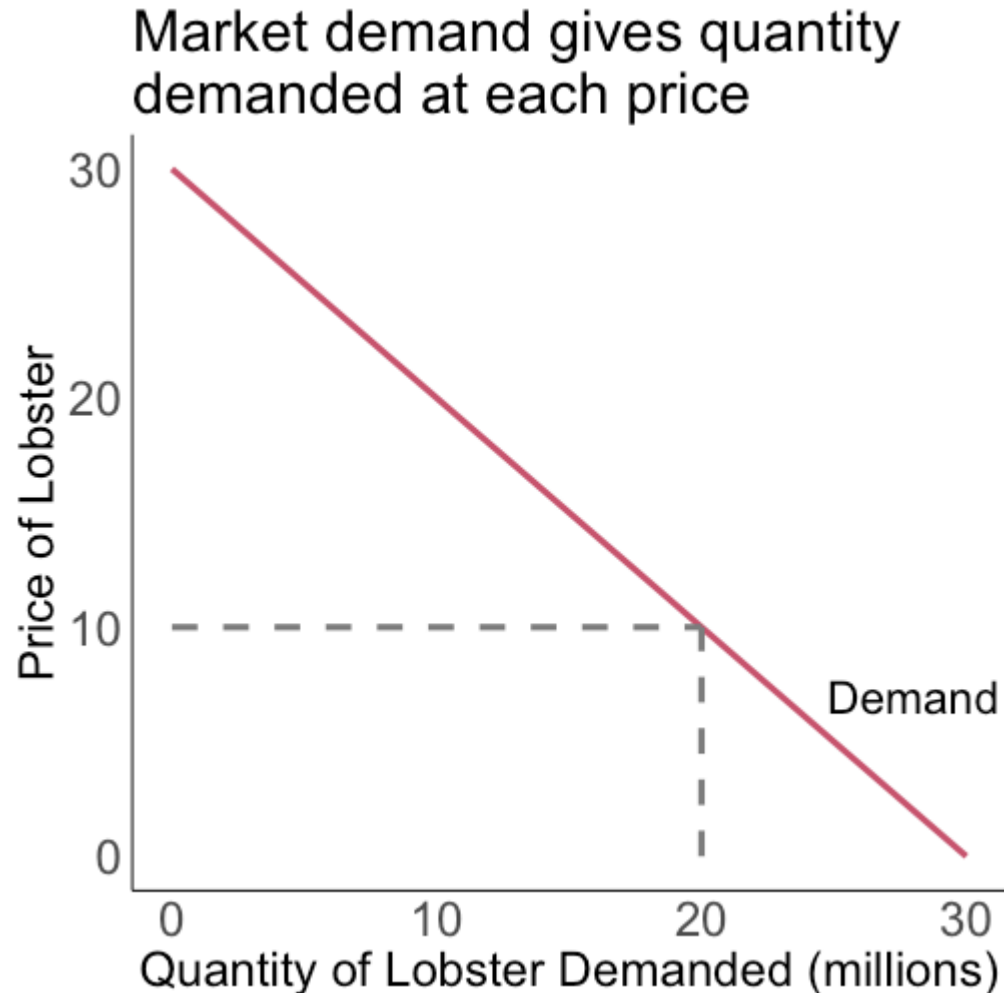
Supply curve: A schedule or graph showing the quantity of a good that sellers wish to sell at each price; it gives us the marginal willingness to accept or the marginal cost

Market demand



Market demand is aggregated from all individual demand curves

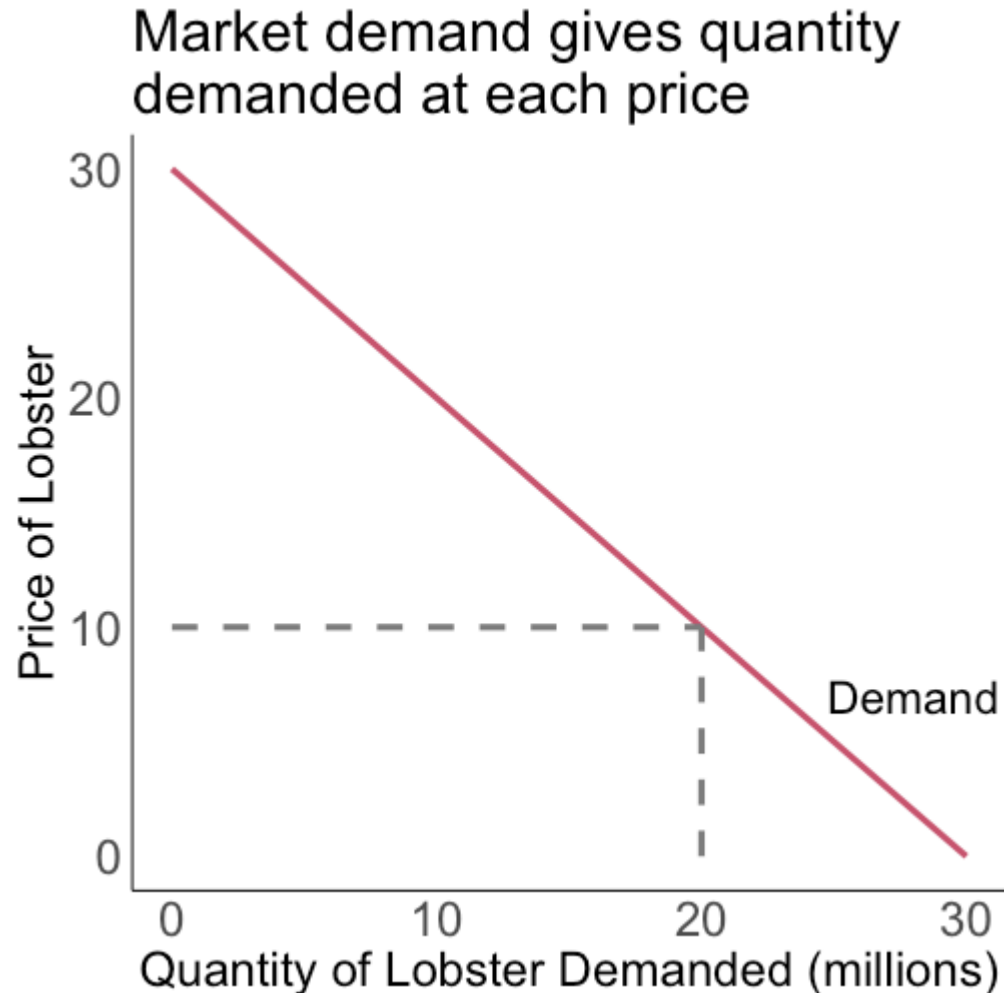
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Vertical interpretation: if buyers are buying 20 million lobsters, the marginal buyer is willing to pay at most \$10

Market demand slopes down

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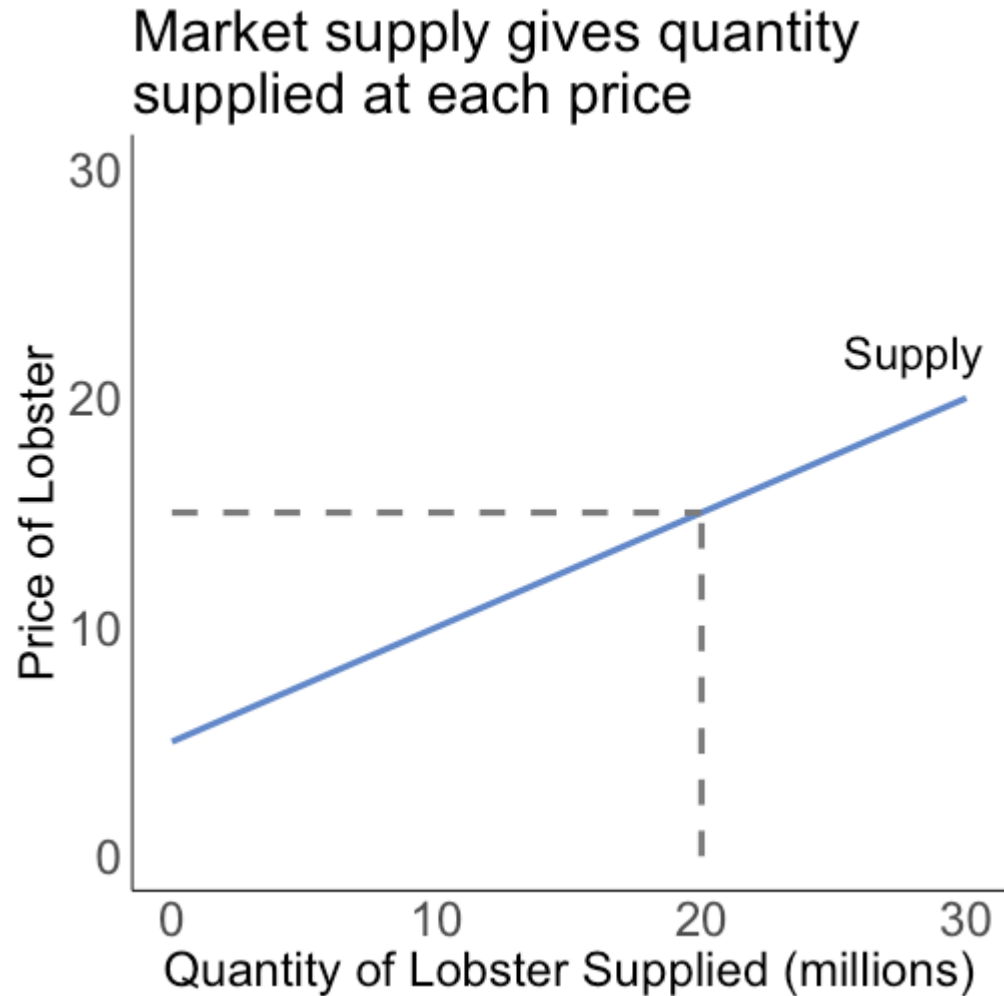
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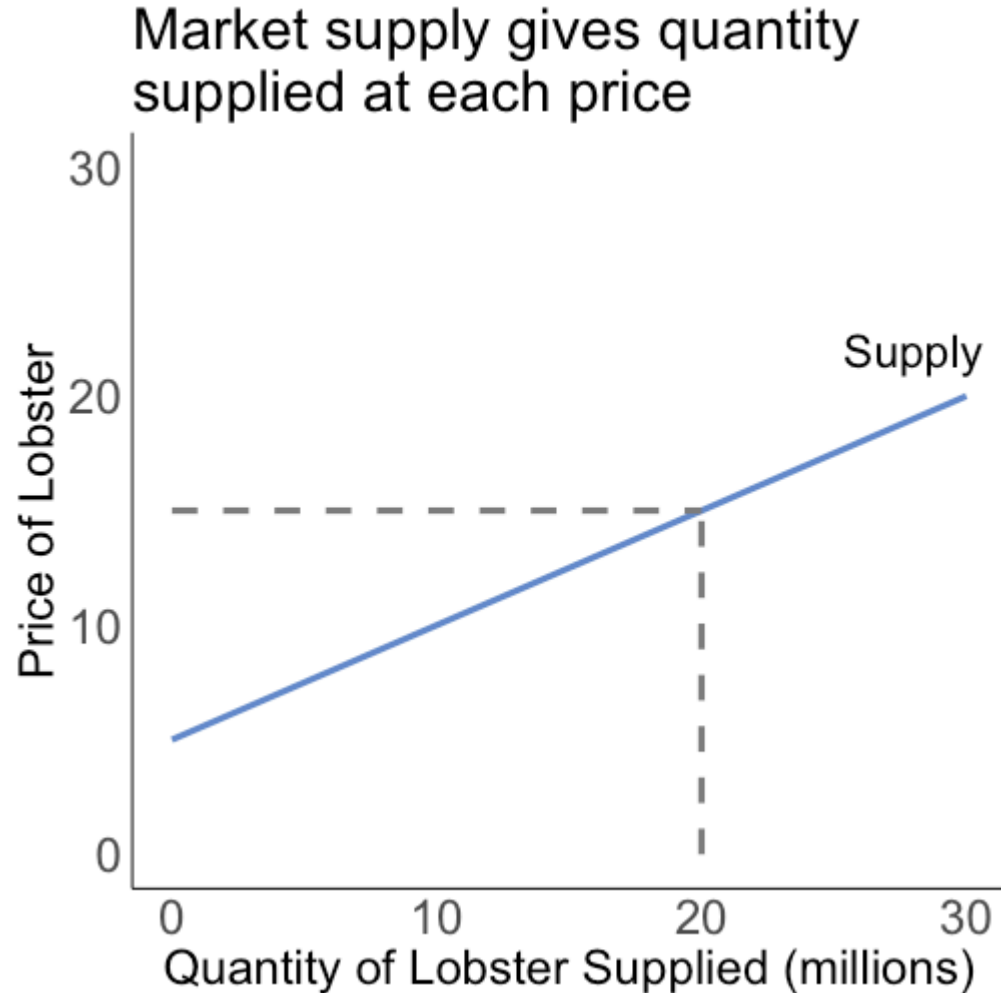
This is the **income effect**: if the price of pizza goes up, we have a lower real budget

Market supply



Market supply is aggregated from all individual supply/MC curves

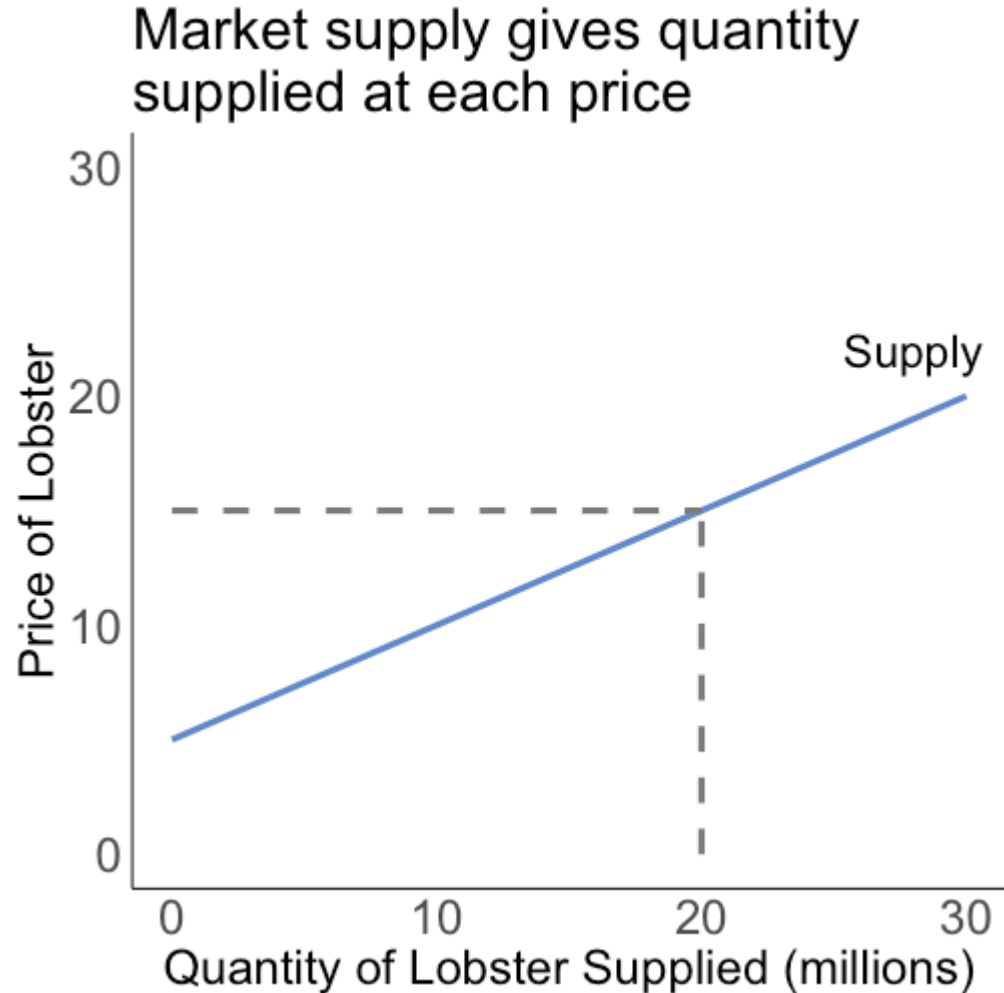
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Vertical interpretation: if sellers are selling 20 million lobsters, the marginal cost of the last lobster is \$15

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If we produce more of a good, we choose the lowest (opportunity) cost production processes first, higher cost production processes later

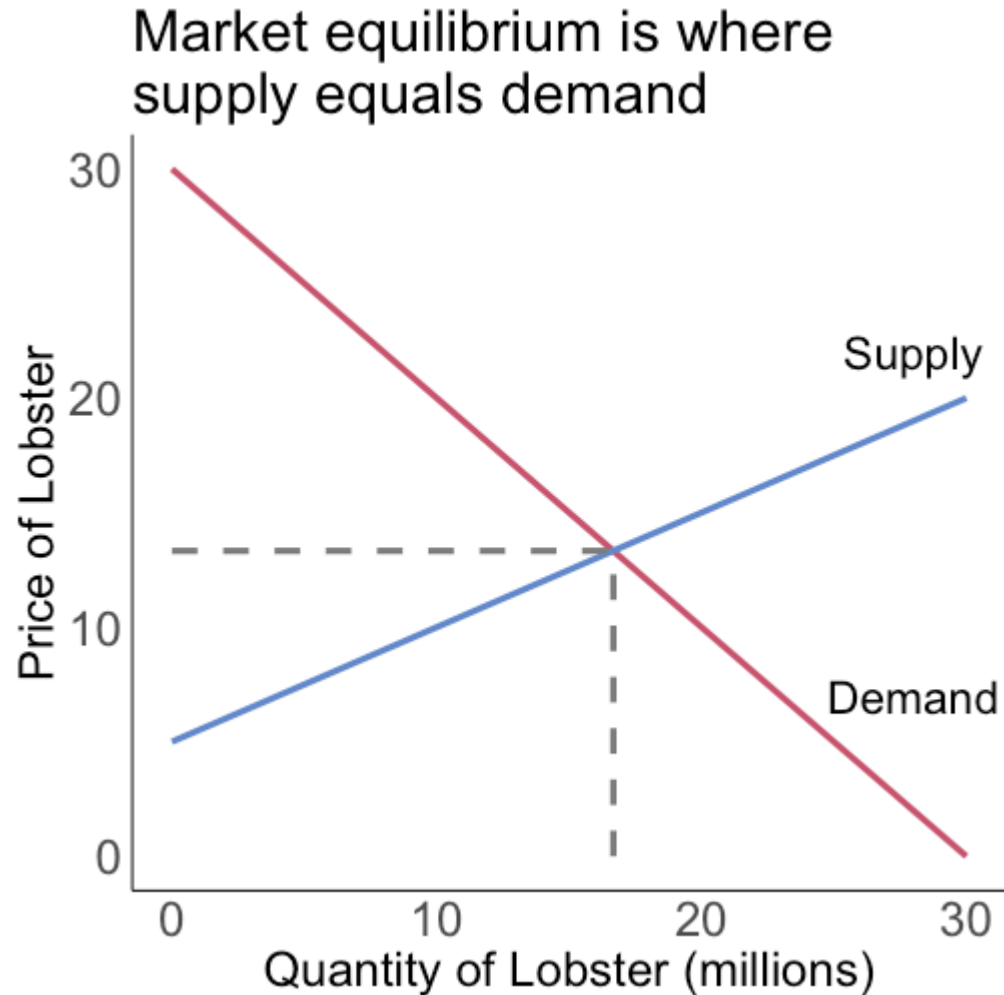
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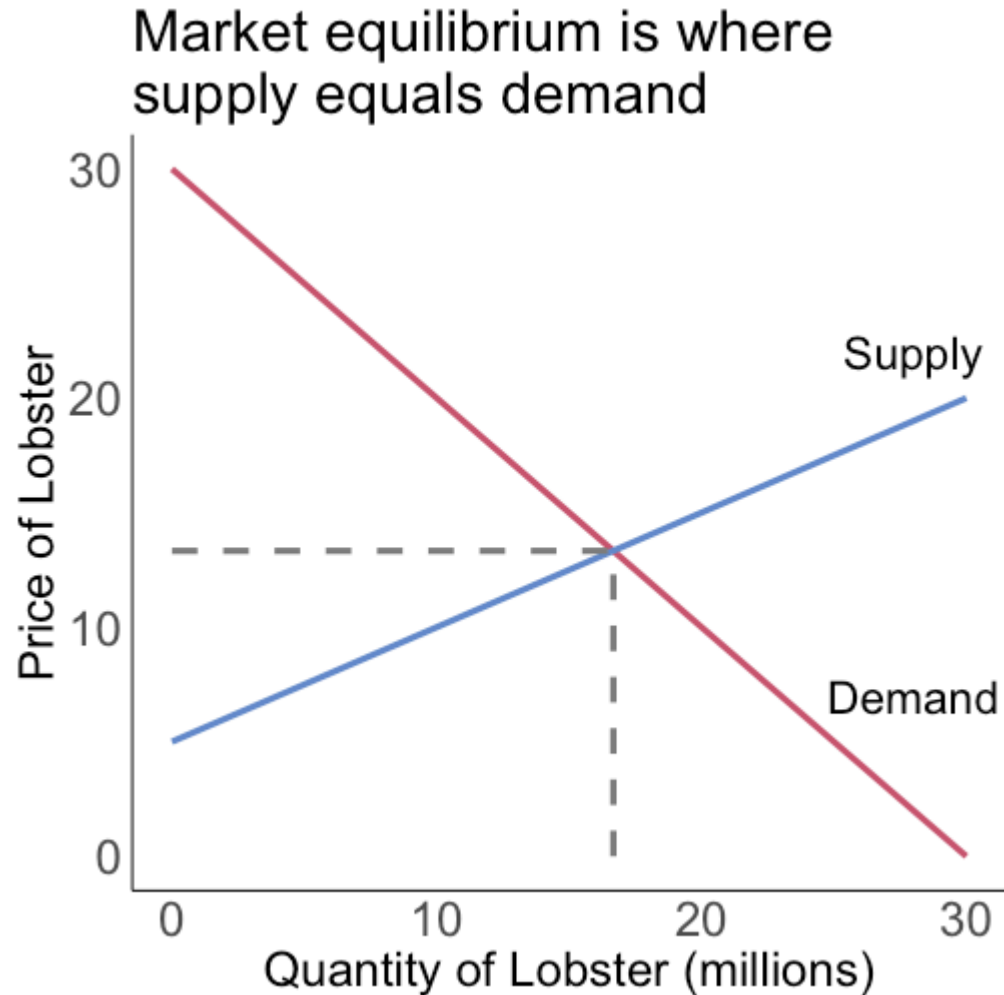
Marginal costs go up as production goes up → producers need higher prices in order to produce more goods

Market equilibrium



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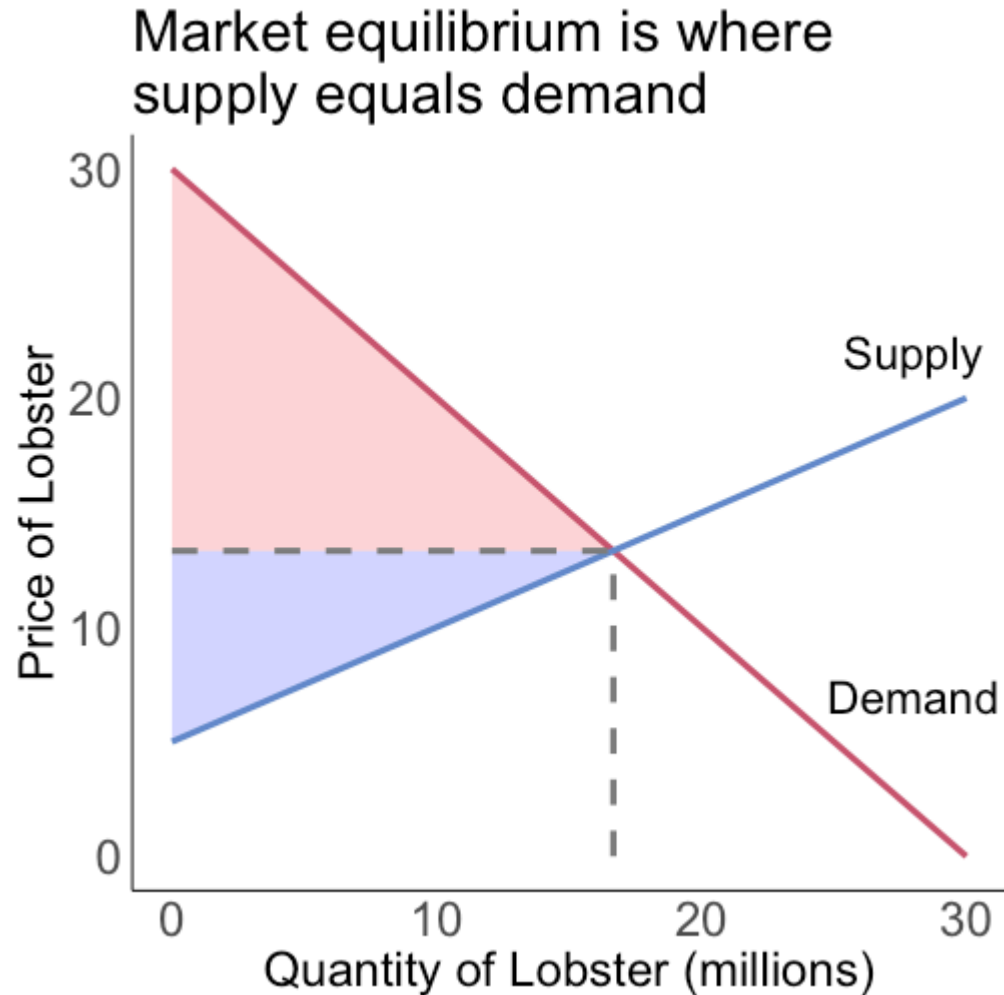
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This gives us the price where the quantity demanded exactly equals the quantity supplied: no shortages, no surpluses

Market equilibrium



Consumer surplus is the difference between willingness to pay (demand) and price

Producer surplus is the difference between price and marginal cost (supply)

Total surplus is the sum of CS and PS

Efficiency

We have two notions of efficiency:

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If you can make one person better off without making anyone else worse off it's called a **Pareto improvement**

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Main takeaway: markets are often a nice way to allocate scarce resources

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3. Complete markets (minimal transactions costs)
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They drive a wedge between private and social marginal cost, or private and social marginal benefit

The key departures in environmental economics

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	Excludable	Non-Excludable
Rival	Private goods (food)	Common-pool resources (fish, timber)
Non-rival	Club goods (parks, netflix)	Public goods (air, national defense)

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Why?

Without the adequate information, buyers or sellers cannot make the choices in their best interest

Efficiency and equity

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- If Elon Musk held **all** the wealth in the world, that would be a Pareto efficient outcome even though it goes against basically all people's notions of equity
- Perfect price discrimination is also Pareto efficient: producers capture all the surplus but it is maximized

Ecological wealth

